Problem 6

Problem 6

Problem 1 *Sketch the graph of a function f satisfying all of the conditions:*

(a) f *is continuous and* odd, $f(0) = 0$,

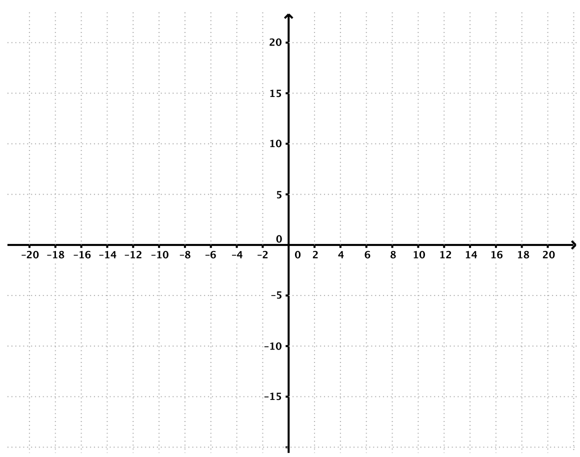
(b) $\lim_{x \to \infty} f(x) = -5$,

(c) $f'(x) > 0$ *on* $(6, \infty)$,

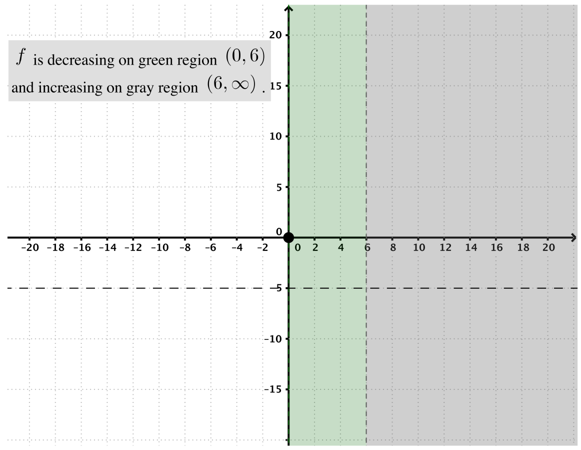
(d) $f'(x) < 0$ *on* $(0, 6)$,

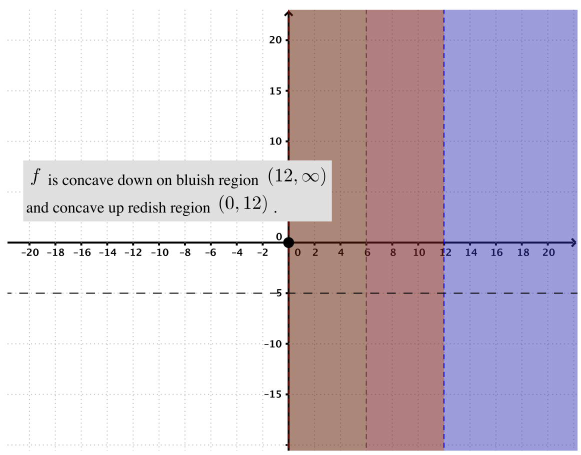
(e) $f''(x) > 0$ *on* $(0, 12)$, *and*

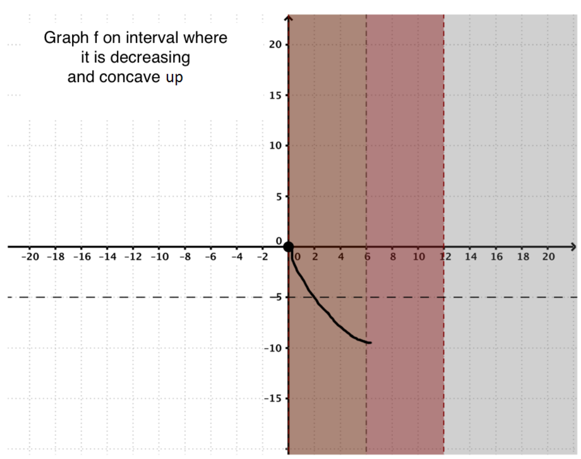
(f) $f''(x) < 0$ *on* $(12, \infty)$.

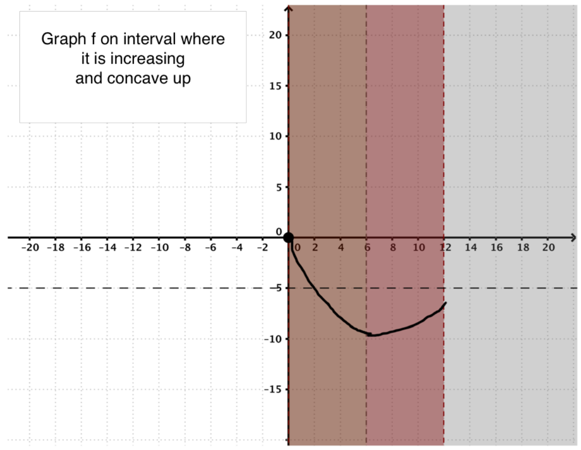


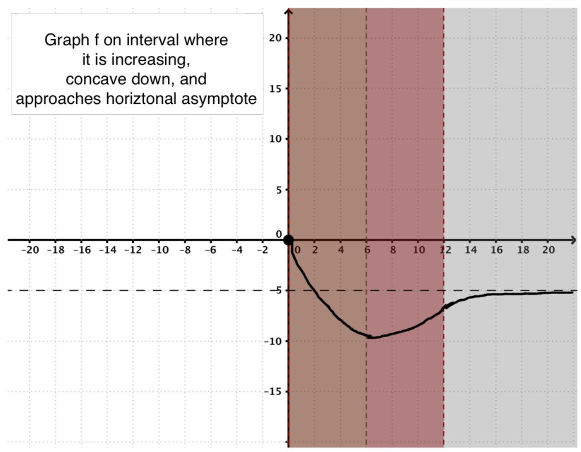
Explanation.

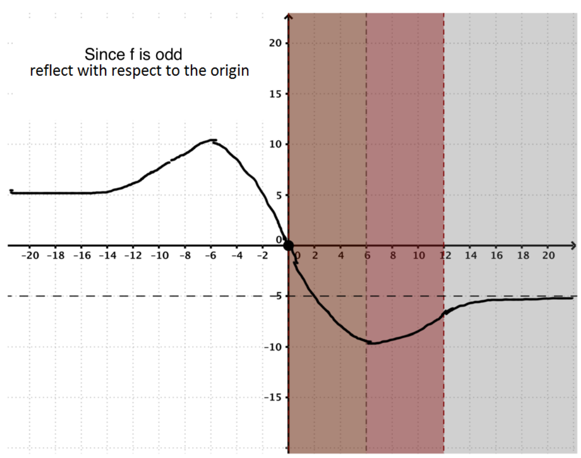

f is decreasing on green region $(0, 6)$
and increasing on gray region $(6, \infty)$.


f is concave down on bluish region $(12, \infty)$
and concave up redish region $(0, 12)$.


Graph f on interval where
it is decreasing
and concave up


Graph f on interval where
it is increasing
and concave up
20
15
10
5
0
-5
-10
-15
-20 -18 -16 -14 -12 -10 -8 -6 -4 -2
0 2 4 6 8 10 12 14 16 18 20


Graph f on interval where
it is increasing,
concave down, and
approaches horiztonal asymptote
20
15
10
5
0
-5
-10
-15
-20 -18 -16 -14 -12 -10 -8 -6 -4 -2
0 2 4 6 8 10 12 14 16 18 20


Since f is odd
reflect with respect to the origin
20
15
10
5
0
-5
-10
-15
-20 -18 -16 -14 -12 -10 -8 -6 -4 -2
0 2 4 6 8 10 12 14 16 18 20